

DEMO UNIVERSITY
END SEMESTER EXAMINATION
DEC 2024

Date : 10/12/2024

Time : 3 Hours

Programme : B.Tech

Branch : Computer Science Engineering

Semester : III

Subject : Database Management System (DBMS)

Course Code : CSE201

Max. Marks : 70

Roll No. : 590017501

Candidate Code : 220101

Student Name : Tanvi Kaur

Note: Attempt all questions. All questions carry equal marks.

Q2. (a) Explain the ACID properties of a transaction. 1

Ans:

1. Atomicity - All operations of a transaction are completed successfully or none.
2. Consistency - A transaction brings the database from one consistent state to another.
3. Isolation - Transactions occur independently without interference.
4. Durability - Once a transaction is committed, changes are permanent even in case of failure.

(b) What is normalization? Explain 1NF and 2NF. 1

Ans: Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

1NF - Each column should contain atomic (indivisible) values and each record must be unique.

2NF - Must be in 1NF and every non-key attribute must be fully functionally dependent on the primary key.